

Distinct PKC-mediated posttranscriptional events set cytokine production kinetics in CD8⁺ T cells

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Effective T cell responses against invading pathogens require the concerted production of three key cytokines: TNF- α , IFN- γ , and IL-2. The cytokines functionally synergize, but their production kinetics widely differ. How the differential timing of expression is regulated remains, however, poorly understood. We compared the relative contribution of transcription, mRNA stability, and translation efficiency on cytokine production in murine effector and memory CD8⁺ T cells. We show that the immediate and ample production of TNF- α is primarily mediated by translation of preformed mRNA through protein kinase C (PKC)-induced recruitment of mRNA to polyribosomes. Also, the initial production of IFN- γ uses translation of preformed mRNA. However, the magnitude and subsequent expression of IFN- γ , and of IL-2, depends on calcium-induced *de novo* transcription and PKC-dependent mRNA stabilization. In conclusion, PKC signaling modulates translation efficiency and mRNA stability in a transcript-specific manner. These cytokine-specific regulatory mechanisms guarantee that T cells produce ample amounts of cytokines shortly upon activation and for a limited time.

T cells | cytokine production | PKC | mRNA stability | translation efficiency

Effective CD8⁺ T cell responses require the production of three key cytokines: tumor necrosis factor alpha (TNF- α), IFN gamma (IFN- γ), and interleukin-2 (IL-2). Intriguingly, T cells release these cytokines in a serial fashion and not simultaneously (1, 2), reflecting their consequential functions. The immediate production of TNF- α induces apoptosis of infected cells and promotes leukocyte infiltration by regulating chemokine release and vascular permeability (3, 4). The slightly later-produced IFN- γ blocks microbial replication and potentiates innate immune responses by directly activating macrophages (5) and recruiting neutrophils (6). The survival and proliferation signals provided by IL-2 are required at later stages of activation. Importantly, all three cytokines have adverse effects when aberrantly produced, which results in immunopathology for TNF- α and IFN- γ (7, 8), and in activation-induced cell death of T cells for IL-2 (9).

Several mechanisms drive cytokine production upon T cell activation. The transcription factors NFAT, AP-1, and NF- κ B are engaged downstream of Ca²⁺ signaling, protein kinase C (PKC), and MAP-kinase signaling (10–13). PKC, MAPK, and PI3K/Akt signaling can also control the turnover of cytokine mRNAs (14–16), whereas mTOR is critical to drive the translation of mRNA into protein (17, 18). How these regulatory networks feed into sequential cytokine production, however, is yet to be determined. We hypothesized that each cytokine requires custom-made regulatory pathways to obtain appropriate antimicrobial responses without inducing severe side effects. Here, we show that the timing of these three cytokines is determined by differential control of transcription, mRNA stability, and translation. Importantly, PKC signaling orchestrates this timing in a cytokine-specific manner.

Results

Sequential Production of Cytokines in T Cells. To determine the kinetics of cytokine release, we activated OT-I T cells for 20 h with OVA_{257–264} peptide/CD80 expressing cells (17). Activated T cells were removed from the antigen after 20 h and rested for 3–13 d in

the presence of IL-7. We then reactivated these effector T cells with OVA_{257–264} peptide in the presence of brefeldin A (BrdA) to block protein secretion and to measure the intracellular cytokine levels of TNF- α , IFN- γ , and IL-2.

TNF- α was detectable within 30 min of stimulation and reached its plateau within the first 2 h, whereas IFN- γ and IL-2 were measured from 1 h and 2 h of activation onwards (Fig. 1A). Protein levels of all cytokines steadily increased until 6 h (Fig. 1A). Whereas this experimental setup displays the maximal capacity of cytokine production, it cannot distinguish whether cytokines are continuously or transiently generated. Therefore, we added BrdA to activated T cells at later time points, for a maximum of 2 h (Fig. 1B). More than 90% of T cells produced TNF- α during the first hours, which dropped to 20% in the last 2 h of stimulation (Fig. 1B). IL-2 production also declined in the last 2 h, whereas IFN- γ production was robust over time (Fig. 1B) (1, 2). Of note, the kinetics of cytokine production was independent of culture conditions with different cytokines, i.e., IL-7, IL-2, or IL-15 (Fig. S1A). Furthermore, the low-affinity T4 variant of OVA_{257–264} peptide (aa: SIITFEKL) (19) only altered the magnitude of response but not the kinetics (Fig. S1A).

When T cells produce two or more cytokines, they represent the most potent effector cells against pathogens (20, 21). We found that the overall percentage of cytokine-producing T cells was constant, but the production profile evolved over time (Fig. 1C and Fig. S1B). TNF- α dominated the first 2 h (91 $\pm$ 4%, of which 54 $\pm$ 5% produced TNF- α alone, and 37 $\pm$ 6% in combination with IFN- γ and/or IL-2). At 4 h, most T cells (90 $\pm$ 2%) produced at least two cytokines. At 6 h, the T cell response shifted toward IFN- γ production (90 $\pm$ 5%, of which 48 $\pm$ 6% produced IFN- γ alone, and 51 $\pm$ 5% together with TNF- α and/or IL-2; Fig. 1C and Fig. S1B).

Also, memory T cells rapidly produce cytokines upon reinfections (22). Indeed, naturally occurring CD44^{hi} memory-like OT-I T cells that can protect from reinfections (23) showed similar production kinetics as *in vitro*-generated effector T cells, irrespective of antigen

Significance

A hallmark of effective T cell responses against invading pathogens is the production of effector molecules. Effector molecules such as cytokines not only kill infected cells, but they also recruit innate immune cells and alter the environment to optimally respond to microbial infections. However, because effector molecules are very potent, they can pose a health risk. When produced excessively or when generated at the wrong time, this aberrant production can induce immunopathology. Here, we uncover that each cytokine employs a private regulation of transcriptional and posttranscriptional events to guarantee appropriate production kinetics.

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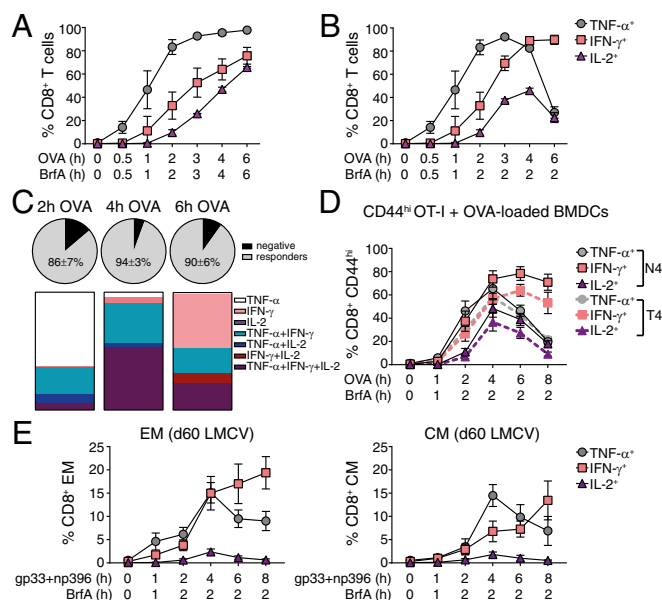


Fig. 1. TNF- α , IFN- γ , and IL-2 follow individual production kinetics. (A and B) Cytokine production of resting CD8⁺ OT-I T cells stimulated with 100 nM OVA peptide. Brefeldin A (BrfA) was present throughout the stimulation (A) or added for the last 0.5, 1, or 2 h of activation (B). (C) Cytokine profile analysis of T cells activated for 2, 4, or 6 h from B. (Upper) Percentage of T cells that produce at least one cytokine. (Lower) Cytokine production of responding T cells. Pooled data from three independently performed experiments (mean $\pm$ SD; $n = 7$ mice). (D) FACS-sorted CD44^{hi} memory-like OT-I T cells activated with peptide-loaded bone marrow-derived dendritic cells. N4, SIINFEKL; T4, SIITFEKL ($n = 5$ mice). (E) Effector memory (EM, Left) and central memory (CM; Right) T cells from LCMV-infected mice (d60) were reactivated with GP33 and NP396 peptide ($n = 5$ mice).

affinity (Fig. 1D). Importantly, effector memory (EM) and central memory (CM) T cells from lymphochoriomeningitis virus (LCMV)-infected mice (d60) also rapidly produced TNF- α and IFN- γ upon stimulation with the cognate GP33 and NP396 peptides, and IL-2 production was delayed and of short nature (Fig. 1E). Different cytokine production kinetics is thus a general feature of effector and of memory T cells.

Preformed *Tnfa* and *Ifng* mRNA Drives Rapid Protein Production. We next interrogated what determines the differential onset of cytokine production. Rapid protein synthesis can be initiated from preformed mRNA. Interestingly, at all culture conditions tested, resting CD8⁺ T cells contained more *Tnfa* mRNA than *Ifng* mRNA, and *Il2* mRNA was almost undetectable (Fig. S1C and Fig. S2A). To examine whether preformed mRNA can drive the rapid TNF- α production, we pretreated T cells with actinomycin D (ActD) to block de novo transcription, or with cycloheximide (CHX) to block de novo translation before activation. CHX significantly blocked the production of TNF- α , but ActD failed to do so. The percentage of TNF- α -producing T cells remained 76 $\pm$ 15% of the control at 30 min of stimulation (Fig. S2B). Thus, CD8⁺ T cells primarily use translation of preformed mRNA to produce TNF- α . Interestingly, preformed *Ifng* mRNA also allows for rapid protein production. After 1 h of activation, 42 $\pm$ 4% of IFN- γ -producing T cells remained in the presence of ActD (Fig. S2C). IL-2, however, depended almost exclusively on de novo transcription. Only 13 $\pm$ 4% T cells remained IL-2⁺ with ActD (Fig. S2D).

Bona fide memory T cells specific for herpes simplex virus (HSV) and LCMV-specific memory T cells (24) also contain preformed *Tnfa* and *Ifng* mRNA, but not of *Il2*, in almost all memory T cell subsets (Fig. 2A). Similarly, CD44^{hi} memory-like T cells express significant *Tnfa* and *Ifng* levels (Fig. 2B). Importantly, both CD44^{hi} T cells and LCMV-specific EM and CM T cells use preformed *Tnfa*

and *Ifng* mRNA for cytokine production, as shown by the negligible effect of ActD on their protein levels (Fig. 2C and D). Of note, the levels of preformed *Ifng* mRNA were higher in memory than in effector T cells (compare Fig. 2A and B with Fig. S2A), which corresponds with higher use of preformed mRNA (Fig. 2C and D) and a more similar onset of production with TNF- α (Fig. 1D and E). Again, IL-2 production primarily depended on de novo transcription (Fig. 2C and D). Combined, our data demonstrate that TNF- α , IFN- γ , and IL-2 follow different production kinetics in effector and in memory CD8⁺ T cells, which is at least in part determined by the levels of preformed mRNA available for rapid translation.

PKC Signaling Contributes to Translation Initiation. Translation of preformed *Tnfa* and *Ifng* mRNA into protein suggests a swift engagement of cytokine mRNA into polyribosomes, a process that is determined by the availability of the eukaryote initiation factor 4E (eIF4E) (25, 26). eIF4E is captured by hypophosphorylated eIF4E-binding proteins (4E-BPs). Phosphorylation of 4E-BP by mTOR releases eIF4E to bind mRNA 5'-cap structures to facilitate the formation and scanning of the preinitiation scanning complex, which allows ribosomes to assemble at start codons (26). Indeed, resting T cells primarily expressed the unphosphorylated α -isoform and the partially phosphorylated β -isoform of 4E-BP1 that both repress translation (Fig. 3A) (25). Stimulation with OVA peptide rapidly decreased the unphosphorylated α -isoform and induced the fully phosphorylated, inactive γ -isoform of 4E-BP1 (Fig. 3A).

We and others showed that mTOR activity drives the translation of cytokines (17, 27). In line with that, preincubating T cells with rapamycin to inhibit mTOR blocks 4E-BP1 phosphorylation (Fig. 3B). TCR-mediated PI3K/Akt signaling phosphorylates mTOR and its downstream target, the ribosomal protein S6 (Fig. 3C and Fig. S3A). This was effectively blocked when T cells were pretreated with the pan-PI3K inhibitor ZTSK474 (Fig. 3C and Fig. S3A). TCR triggering, however, also engages other crucial signaling pathways, such as the Ca²⁺ flux that activates calcineurin (28), and PKC (29). We therefore questioned whether Ca²⁺ flux and/or PKC signaling also mediate mTOR activity and, thus, translation. Inhibiting calcineurin with cyclosporine A (CsA) did not affect mTOR and S6 phosphorylation (Fig. 3C and Fig. S3A). In contrast, blocking PKC with LY-333531 (specific inhibitor of PKC- β isoform) or sotrastaurin (pan-PKC inhibitor) before activation significantly impaired mTOR and S6 phosphorylation (Fig. 3C and Fig. S3A). Thus, TCR-dependent activation of PKC can enhance mTOR activity.

PKC promoted mTOR and S6 phosphorylation independently from PI3K signaling. Stimulating T cells with phorbol 12-myristate 13-acetate (PMA) alone to directly engage PKC sufficed for mTOR and S6 phosphorylation, whereas promoting the intracellular Ca²⁺ influx with ionomycin had no effect (Fig. S3B). Furthermore, pretreating cells with PKC inhibitors LY-333531 and sotrastaurin effectively blocked mTOR and S6 phosphorylation upon PMA/ionomycin stimulation (Fig. 3D and Fig. S3C). Importantly, the PI3K inhibitor ZTSK474 and the calcineurin inhibitor CsA failed to do so, demonstrating that PKC-mediated activation of mTOR was independent from PI3K and Ca²⁺ influx (Fig. 3D). T cell activation with PMA alone was also sufficient to induce the hyperphosphorylated inactive γ -isoform of 4E-BP1 (Fig. S3D). Again, this shift was blocked by rapamycin, and by LY-333531 or sotrastaurin, but not by CsA (Fig. 3E). In accordance with a more diverse signal upon TCR triggering and a possible redundancy between the PI3K and the PKC pathway, inhibiting PKC upon OVA peptide stimulation did not reduce the migration of 4E-BP1 toward its γ -isoform (Fig. S3E). Altogether, our data show that PKC signaling converges with the PI3K pathway to regulate mTOR activity and to initiate mRNA translation of activated T cells.

Differential Role of PKC and Ca²⁺ in Cytokine Production. We next questioned how PKC and Ca²⁺ signaling contribute to cytokine production. PMA/ionomycin activation induced cytokine production with similar kinetics as antigen stimulation (Fig. 4A and

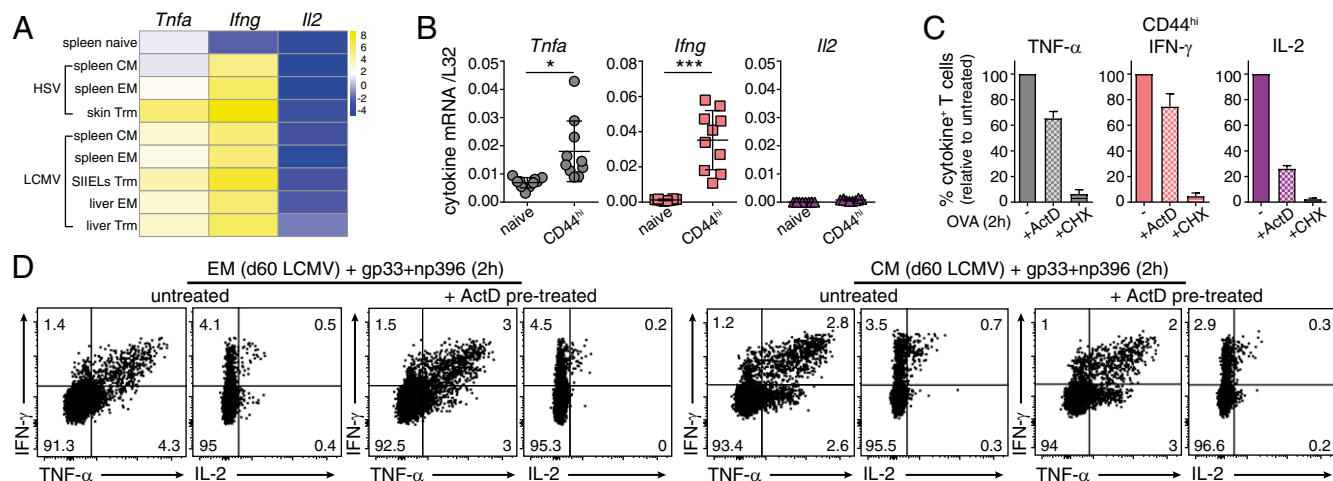


Fig. 2. Preformed cytokine mRNA promotes early T cell responsiveness. (A) Heatmap of the relative expression (Z scores) of *Tnfa*, *Ifng*, and *Il2* mRNA (ref. 24; GSE70813). (B) *Tnfa*, *Ifng*, and *Il2* mRNA levels of CD8⁺ CD44^{low} naive T cells and CD8⁺ CD44^{hi} memory-like T cells ($n = 10$ mice) (paired Student's t test; $^{**}P < 0.005$; $^{***}P = 0.0001$). (C) Relative percentage of TNF- α , IFN- γ , and IL-2 producing CD8⁺ CD44^{hi} T cells pretreated with 1 μ M ActD or 10 μ M CHX for 30 min before 2 h stimulation with OVA peptide ($n = 6$ mice). (D) Cytokine production of EM and CM T cells from LCMV-infected mice pretreated with ActD before 2 h GP33 and NP396 peptide stimulation. Representative of five mice from two independently performed experiments.

Fig. S4 A and B). Intriguingly, however, PKC and Ca²⁺ signaling differentially contributed to the individual cytokine production. Ionomycin only marginally supported TNF- α , while IFN- γ largely depended on Ca²⁺ signaling. In sharp contrast, PMA stimulation alone had limited effects on IFN- γ , but greatly promoted TNF- α production (Fig. 4A and Fig. S4A). This finding was confirmed with PKC and calcineurin inhibitors in PMA/ionomycin-stimulated and antigen-stimulated T cells (Fig. 4B and C). This indicates that PKC and Ca²⁺ signaling have divergent roles for cytokine production also when all TCR-mediated signaling nodes are present (Fig. 4C and Fig. S4D).

Strikingly, cytokine production did not necessarily correlate with mRNA levels. PMA alone induced substantial production of TNF- α protein in effector and CD44^{hi} memory-like T cells (Fig. 4A and D and Fig. S4A), but this did not correspond with significant changes in *Tnfa* mRNA (Fig. 4D and Fig. S4E). Increases of transcript levels became apparent only upon combined PMA/ionomycin activation (Fig. 4D). In contrast, *Ifng* mRNA already increased upon single stimulation with ionomycin and was further boosted by the combination of PMA/ionomycin (Fig. 4E and Fig. S4E). Thus, *Ifng* but not *Tnfa* transcript levels match with the protein levels.

We next questioned how PKC and Ca²⁺ signaling influenced the translation efficiency of *Tnfa* and *Ifng* mRNA. We fractionated polyribosomes from cytoplasmic extracts of activated T cells by sucrose gradient centrifugation, and determined the distribution of *Tnfa* and *Ifng* transcripts between free and polyribosome-bound fractions of increasing density (Fig. 4G and H and Fig. S4F). A large fraction of *Tnfa* mRNA remained as free-mRNA upon T cell activation with ionomycin alone (Fig. 4G). PMA activation, however, shifted *Tnfa* from free-mRNA toward polyribosome-bound fractions. These findings are fully compatible with the capacity of PKC to phosphorylate 4E-BP1 (Fig. 3E) and to produce TNF- α protein (Fig. 4A–D). Combined PMA/ionomycin activation resulted in further shifts of *Tnfa* mRNA from light to heavy polyribosomes (Fig. 4G), indicating that Ca²⁺ signaling—although not sufficient to initiate *Tnfa* translation—can contribute to protein production by increasing *Tnfa* mRNA levels, and by enhancing polysomal recruitment.

In contrast to *Tnfa*, we found *Ifng* already in polyribosomes in ionomycin-activated cells (Fig. 4H). PMA stimulation, however, was more effective at evoking high-density ribosome loading of *Ifng* mRNA (Fig. 4H). Because ionomycin also induced de novo transcription of *Ifng* (Fig. 4E and Fig. S4E), we postulate that newly

synthesized *Ifng* mRNA is immediately engaged by ribosomes, resulting in the observed protein production.

IL-2 production required both PMA and ionomycin stimulation for measurable increases of mRNA levels and cytokine production (Fig. 4A and F). As with *Ifng* mRNA, *Il2* mRNA always associated with polyribosomes, which was further enhanced by combined PMA/ionomycin stimulation (Fig. 4I). Nevertheless, *Il2* mRNA levels at ionomycin activation alone are possibly too low to allow for detectable protein production (Fig. 4A and F). Thus, IL-2 requires both PKC and Ca²⁺ signaling for detectable protein levels.

Signal Strength Dictates Protein Production in a Cytokine-Specific Manner. The levels of cytokines produced by T cells depend on costimulatory signals, on the affinity, and the amount of antigen (30, 31). We stimulated T cells with low (0.1 nM), intermediate (1 nM), and high (100 nM) OVA peptide concentration to determine the effect on cytokine mRNA expression and protein levels. OVA^{low} did

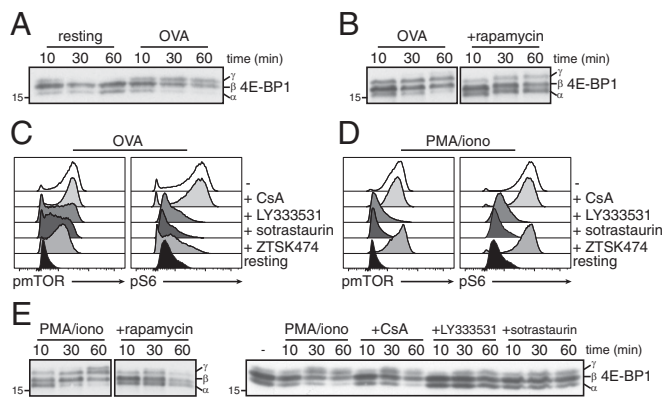


Fig. 3. PKC signaling promotes translation initiation. (A and B) 4E-BP1 expression in resting and OVA peptide-stimulated T cells (A), and in OVA-stimulated T cells pretreated for 30 min with rapamycin (B). (C and D) pmTOR and pS6 levels 30 min after T cell activation with OVA peptide (C), or with PMA/ionomycin (D). Indicated inhibitors were added 30 min before T cell activation. (E) 4E-BP1 expression of PMA/ionomycin activated T cells that were pretreated or not with indicated inhibitors. Representative of two independently performed experiments (A, B, and E) and 10 mice from three experiments (C and D).

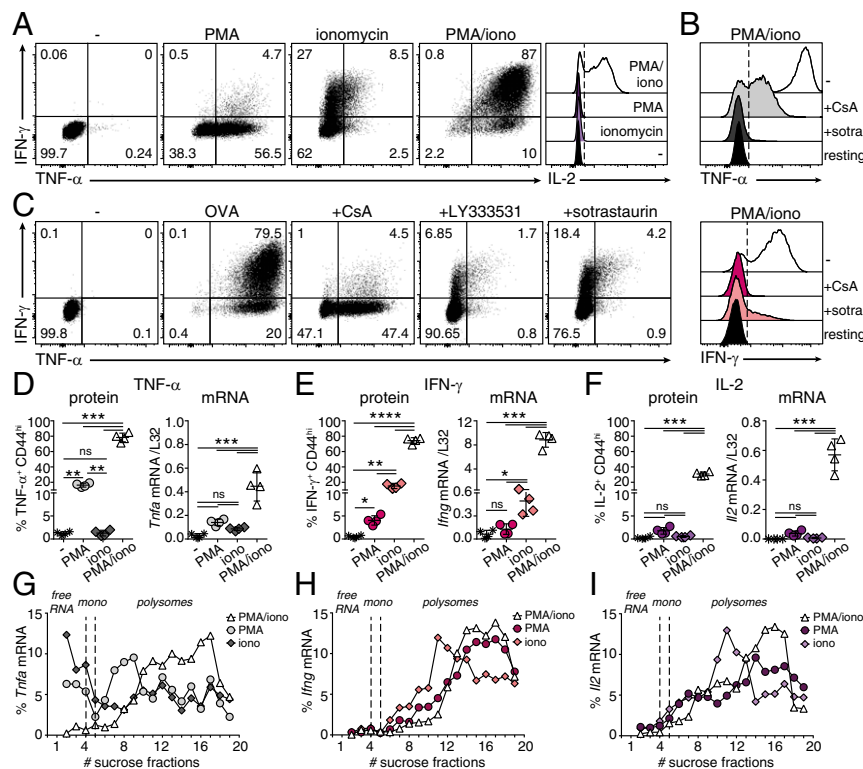


Fig. 4. Role of PKC and Ca^{2+} -dependent signaling in cytokine production. (A) Resting T cells were stimulated or not for 4 h with PMA, ionomycin, or with both (PMA/ionomycin). Numbers in dot plots indicate percentage of $\text{TNF-}\alpha^+$ and/or $\text{IFN-}\gamma^+$ T cells. Histogram depicts IL-2 production of the same sample. (B and C) T cells were pretreated or not with indicated inhibitor before activation with PMA/ionomycin (B), or with OVA peptide (C). Representative of three independently performed experiments. (D–F) $\text{TNF-}\alpha$ (D), $\text{IFN-}\gamma$ (E), and IL-2 (F) protein levels (Left) and mRNA levels (Right) of CD44^{hi} memory-like T cells were determined after 4 h and 2 h of indicated stimulation, respectively ($n = 4$ mice) (one-way ANOVA with Tukey's comparison; * $P < 0.05$; ** $P < 0.005$; *** $P = 0.0005$; **** $P < 0.0001$). (G–I) Polysome fractionation of T cells activated for 2 h with indicated stimuli. *Tnfa* (G), *Ifng* (H), and *Il2* (I) mRNA distribution through all 20 collected fractions. Representative of two independently performed experiments.

not drive protein production of any of the three cytokines. Upon activation with OVA^{int} , a substantial fraction of T cells produced $\text{TNF-}\alpha$ ($48 \pm 14\%$; Fig. 5A), and a minor but significant fraction produced $\text{IFN-}\gamma$ ($21 \pm 3\%$; Fig. 5B). IL-2 was almost undetectable ($2 \pm 1\%$; Fig. 5C), and T cells only produced significant IL-2 levels when activated with OVA^{hi} .

Strikingly, again only activation with OVA^{hi} resulted in elevated *Tnfa* transcript levels, but OVA^{int} failed to induce *Tnfa* mRNA despite the substantial $\text{TNF-}\alpha$ protein production (Fig. 5D). We therefore interrogated how antigen load affected the recruitment of mRNA to polysomes. In T cells activated with OVA^{low} , *Tnfa* was primarily found as free-RNA (Fig. 5D). Importantly, activation with OVA^{int} totally shifted *Tnfa* toward polyribosome-bound fractions. This was not further amplified with OVA^{hi} , suggesting that the translation efficiency could not increase anymore (Fig. 5D). Thus, $\text{TNF-}\alpha$ production depends on recruitment of preformed *Tnfa* mRNA to polyribosomes, which can be further amplified by de novo transcription at high antigen levels.

In sharp contrast, *Ifng* and *Il2* transcript levels directly correlated with protein levels: OVA^{int} and OVA^{hi} significantly increased *Ifng*, as did OVA^{hi} for *Il2* (Fig. 5E and F). Furthermore, *Ifng* and *Il2* almost completely associated with polyribosomes at any condition tested (Fig. 5E and F), again indicating that cytoplasmic *Ifng* and *Il2* mRNA immediately engages with ribosomes. Altogether, we show that mRNA transcription and mRNA translation differentially contribute to the production of $\text{TNF-}\alpha$, $\text{IFN-}\gamma$, and IL-2.

PKC-Signaling Stabilizes *Ifng* and *Il2* Transcripts for Optimal Protein Production. In resting T cells, *Tnfa* mRNA levels are higher than those of *Ifng* and *Il2* (Fig. S4A). However, after 2 h activation with

OVA^{hi} , *Ifng* and *Il2* mRNA levels steeply increase and outnumber those of *Tnfa* mRNA (Fig. 6A), independently of the culture conditions or the antigen affinity (Fig. S5A). Rapid accumulation of transcripts can be promoted by de novo transcription and mRNA stabilization. mRNA stability was associated with increased translation efficiency, including transcripts that contain the regulatory AU-rich elements within their 3'UTR, like *Tnfa*, *Ifng*, and *Il2* (16, 32). We thus questioned whether optimal cytokine production correlated with mRNA stability. T cells activated for 2 h with OVA^{hi} or PMA/ionomycin were treated with ActD over the course of the next 2 h to measure the mRNA decay rate. The half-life of *Tnfa* was unaffected by T cell stimulation irrespective of the stimulus used ($t_{1/2} = \sim 60$ min; Fig. 6B and Fig. S5B), which was in line with its moderate increase in mRNA levels (Fig. 6A). In sharp contrast, *Ifng* and *Il2* mRNA transcripts were substantially stabilized upon T cell activation to $t_{1/2} > 2$ h (Fig. 6C and D and Fig. S5B).

We next studied which signaling pathways drive the stabilization of *Ifng* and *Il2* mRNA. We added PKC or Ca^{2+} -flux inhibitors combined with ActD to T cells that were activated for 2 h with OVA^{hi} or PMA/ionomycin. Again, *Tnfa* half-life was not affected by any inhibitor (Fig. 6B and Fig. S5B). CsA also did not alter the stability of *Ifng*, and only slightly reduced the half-life of *Il2* mRNA to ~ 115 min (Fig. 6C and D and Fig. S5B). However, the half-life of both *Ifng* and *Il2* dropped back to levels of resting T cells when with PKC inhibitor LY-333531 or sotrastaurin (Fig. 6C and D and Fig. S5B). Furthermore, activating PKC with PMA alone was already sufficient to fully stabilize *Ifng* and *Il2* mRNA, while ionomycin failed to do so (Fig. S4C). Thus, PKC is the main driver for *Ifng* and *Il2* stabilization.

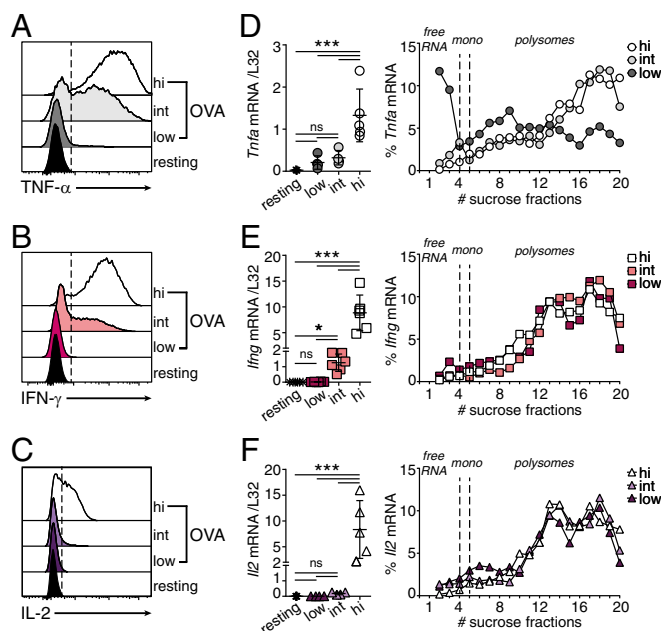


Fig. 5. Signal strength determines cytokine specific pathway for production. (A–C) TNF- α (A), IFN- γ (B), and IL-2 (C) protein production upon 6 h of stimulation with 0.1 nM (low), 1 nM (intermediate), or 100 nM (high) OVA peptide. Representative of four independently performed experiments ($n = 6$ mice). (D–F, Left) *Tnfa* (D), *Ifng* (E), and *Il2* (F) mRNA levels after 2 h of stimulation. Mean $\pm$ SD of three (D and F) and four (E) independently performed experiments [$n = 5$ (D and F); $n = 6$ (E) mice] (one-way ANOVA with Tukey's comparison; * $P < 0.05$; *** $P < 0.0005$). (Right) Polysome fractionation of cytoplasmic *Tnfa* (D), *Ifng* (E), and *Il2* (F) mRNA after 2 h stimulation with OVA peptide. Results are representative of two independently performed experiments.

Of note, *Ifng* mRNA maintained its stability also at later time points, i.e., at 4 h after T cell activation (Fig. 6E and Fig. S5D). In contrast, the half-life of *Il2* at this time point dropped back to levels of resting T cells (Fig. 6F), and *Tnfa* maintained its low $t_{1/2}$ of ~ 60 min at any time point measured (Fig. S5D). The rapid turnover of *Tnfa* and *Il2* mRNA between 4 and 6 h after activation was in line with the profound decrease of protein production at this time point (Fig. 1 and Fig. S4B). Combined, we here show that mRNA stability is indicative for the duration of cytokine production and that PKC signaling herein is key.

Discussion

A potent T cell response requires the coordinated production of TNF- α , IFN- γ , and IL-2. We show that effector and memory T cells use an individual regulatory program for each cytokine to drive production with distinct kinetics. TNF- α production was fast and intense, but transient, and depended on immediate polyribosome recruitment of preformed mRNA. T cells also use preformed mRNA for the early production of IFN- γ . In contrast, the delayed IL-2 production depended almost exclusively on de novo gene transcription. In addition, stabilization of newly synthesized mRNA defined the duration of cytokine production. Prolonged mRNA half-life supported the sustained production of IFN- γ , whereas *Il2* mRNA lost its stability within 4 h concomitant with a gradual loss of protein production.

Although TNF- α , IFN- γ , and IL-2 use private regulatory networks, PKC signaling is crucial for all three cytokines. However, its mode of action is cytokine-specific. PKC drives *Tnfa* translation without stabilizing its mRNA. Conversely, PKC mediates mRNA stability of *Ifng* and *Il2* and enhances the shift of mRNA from light to heavy polyribosomes.

PKC activation promotes mRNA translation through mTOR, and the two mTOR targets S6 and 4E-BP1 that drive protein synthesis (27). Whether PKC acts directly or indirectly on these targets remains to be determined. Direct phosphorylation of S6 and 4E-BP1 by PKC has been reported (33, 34), but PKC could also regulate mTOR and 4E-BP activity through i.e., ERK-dependent mechanisms (35, 36). Irrespective of its mode of action, PKC-induced phosphorylation of 4E-BPs releases eIF4E, allowing the preinitiation scanning complex to assemble at the mRNA cap (25, 26). The predominant role of polysome binding and mRNA translation in regulating TNF- α , and its addition to 4E-BP1 phosphorylation, may in part explain the success of rapamycin and its analogs as immune suppressants, since those inhibitors are capable to directly block 4E-BP phosphorylation (37).

How PKC stabilizes cytokine mRNA is yet to be determined. In Jurkat cells, PKC phosphorylates the RNA-binding protein NF-90, which allows NF-90 to bind and stabilize *Il2* mRNA (15). Similarly, HuR can stabilize *Ifng* mRNA (38), and in vitro data show that PKC can phosphorylate HuR (39). Thus, it is tempting to speculate that HuR and NF-90 promote *Ifng* and *Il2* mRNA stability in primary CD8⁺ T cells in a PKC-mediated manner.

T cells that produce two or more cytokines are the most effective against viruses (20, 21, 40), and their capacity to induce multi-flavored responses defines the efficacy of T cell vaccines (41). Here, we show that the kinetics of cytokine production is a general feature of effector and memory T cells and that it is cell-intrinsic. PKC orchestrates the timing of production of each cytokine by providing a custom-made posttranscriptional regulation. Understanding the mechanisms that drive optimal cytokine production should further

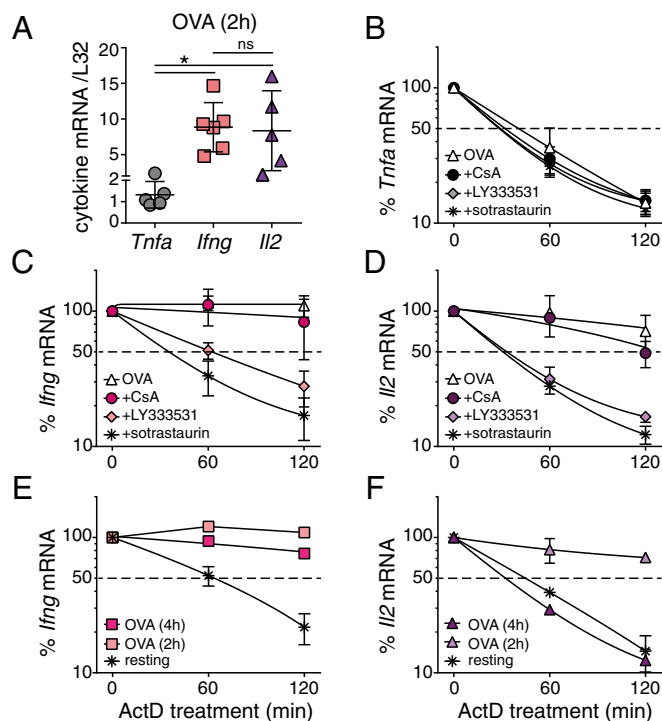


Fig. 6. PKC activation regulates the stability of *Ifng* and *Il2* mRNA. (A) *Tnfa*, *Ifng*, and *Il2* mRNA levels upon 2 h stimulation with OVA^{hi}; data are extracted from Fig. 5 D–F (one-way ANOVA with Tukey's comparison; * $P < 0.05$). (B–F) T cells stimulated with OVA^{hi} for 2 h (B–D), or for 2 h and 4 h (E and F) and then were incubated with ActD alone (OVA), or in combination with CsA, LY-333531, or sotrastaurin. Cells were harvested at indicated time points to determine the mRNA decay of *Tnfa* (B), *Ifng* (C and E), and *Il2* (D and F) ($n = 5$ mice from two independently performed experiments).

help to unravel how effective, multifunctional T cell responses can be generated.

Materials and Methods

Activation and Expansion of Murine Effector and Memory CD8⁺ T Cells. Mice were housed and bred at the Netherlands Cancer Institute (NCI). Experiments are approved by the Experimental Animal Committee of the NCI. OT-I T cells were activated as described (17). Activated T cells were rested for 3–13 d in 10 ng/mL recombinant murine (rm)IL-7, 120 U rmlL-2, or 10 ng/mL rmlL-15 (Peprotech). Splenic CD44^{lo} and CD44^{hi} OT-I T cells, and splenic CD44^{hi} CD62L⁺ and CD44^{hi}CD62L[−] memory T cells from LCMV-Armstrong-infected mice (d60; 10⁵ pfu per mouse, i.p. injection) were FACS-sorted. T cells were stimulated with 100 nM OVA_{257–264} peptide, or its low-affinity T4 variant, with 10 ng/mL PMA, 1 μM ionomycin (Sigma-Aldrich), or with peptide-loaded bone marrow-derived DCs. Brefeldin A was added as indicated. DCs were incubated for 1.5 h with 100 nM peptide and washed twice. LCMV-specific memory T cells were stimulated with 5 μg/mL GP33 and 5 μg/mL NP396 peptide (kind gift of R. Arens, Leiden University Medical Center, Leiden, The Netherlands). Inhibitors were added to T cells 30 min before activation (Table S1).

Flow Cytometry. Intracellular cytokine staining was performed with BD Cytofix/Cytoperm kit (BD Biosciences). pMTOR and pS6 was measured upon 4% PFA fixation and methanol permeabilization. For antibodies, see Table S2. Cells were acquired with LSR Fortessa (BD Biosciences) and analyzed by using FlowJo software (Tree Star, version 10).

Polysome Fractionation. Activated T cells (40 × 10⁶) were washed with PBS containing 100 μg/mL cycloheximide. The cytoplasmic extract was layered on a 17–50% sucrose gradient (42) and centrifuged for 2 h at 178,305 × g (38,000 rpm)

at 4 °C in a SW41Ti rotor (Optima L100XP ultracentrifuge; Beckman Coulter). RNA was isolated by Phenol:Chloroform extraction. Total polyribosome-associated mRNA levels were quantified by RNA Nano Chip assay (Agilent). Relative mRNA distribution was calculated as percent of the sum of mRNA from all obtained fractions. Total mRNA levels were put at 100% for each condition.

Quantitative PCR Analysis. cDNA was synthesized by using SuperScript III Reverse Transcriptase (Invitrogen). Quantitative RT-PCR was performed with SYBR Green on StepOne Plus RT-PCR (Applied Biosystems) (17). Gene expression levels were calculated by the 2^{−ΔΔCt} analysis. Cytokine mRNA half-life was determined with GraphPad Prism 6 by calculating the one-phase exponential decay curve.

Western Blot. Cell lysates (1 × 10⁶ cells/sample) were prepared by standard procedures. Proteins were separated on a 15% SDS/PAGE and transferred onto a nitrocellulose membrane by iBlot (Thermo). Anti-4E-BP1 (Cell Signaling Technology) (43), followed by goat anti-rabbit-HRP secondary antibody (Dako), was used.

Statistical Analysis. Results are expressed as mean ± SD. Data were analyzed with GraphPad Prism6. Unpaired or paired two-tailed Student's *t* test was used when comparing two groups, one-way ANOVA test with Tukey's multiple comparison, or Dunnett's multiple comparison with control group was used when comparing >2 groups. *P* values of <0.05 were considered statistically significant.

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